

# Math 122: Exam 1 - Practice 2

**Student Name:** \_\_\_\_\_

**Instructions:**

- Show all work: Full credit requires complete, correct steps and proper math notation.
  - Points: Each question is worth 10 points.
  - Formula Sheet: Permitted (formulas only; no worked examples).
  - Calculators: Basic, non-graphing calculators allowed.
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**Problem 1**

Find the exact arc length of the curve over the interval from  $x = 0$  to  $x = 1$ :

$$y = (2x + 1)^{3/2}$$

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**Problem 2**

Find the limit:

$$\lim_{x \rightarrow 0} \frac{\tan(2x)}{\ln(1+x)}$$

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**Problem 3**

Find the limit:

$$\lim_{x \rightarrow +\infty} x^2 e^{-2x}$$

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**Problem 4**

Find the limit:

$$\lim_{x \rightarrow 0^+} (1 + 2x)^{1/x}$$

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**Problem 5**Find  $\frac{dy}{dx}$ :

$$y = \cos^{-1} \left( \frac{1}{x^2} \right)$$

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**Problem 6**Find  $\frac{dy}{dx}$ :

$$y = \sin^{-1} x \cdot \cos^{-1}(x)$$

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**Problem 7**

Evaluate the integral:

$$\int \frac{dx}{1 + 9x^2}$$

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**Problem 8**

Evaluate the integral:

$$\int \frac{x \, dx}{\sqrt{1 - x^4}}$$

**Problem 9**

Evaluate the definite integral:

$$\int_0^1 \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$$

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**Problem 10**

Evaluate the integral:

$$\int \frac{e^x}{1+e^{2x}} dx$$

— END OF EXAMINATION —